

HDOC - Day 15

Check Whether All Digits Are Even or Prime

Algorithm

1. Start. 2. Read a positive integer. 3. Display the menu. 4. For option 1, examine every digit; if any digit is odd, mark the condition false. 5. For option 2, examine every digit; only 2, 3, 5 and 7 are prime digits. If any other digit occurs, mark the condition false. 6. Display the result. 7. Repeat until Exit. 8. Stop.

Test Case Table

TC	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	12345; choice 1	All digits not even	All digits not even	Pass
2	24680; choice 1	All digits are even	All digits are even	Pass
3	2357; choice 2	All digits are prime	All digits are prime	Pass
4	2379; choice 2	All digits are not prime	All digits are not prime	Pass

C Program

```
#include <stdio.h>
int main() {
    int n, choice, temp, d, allEven, allPrime;
    do { printf("Enter a positive integer: "); scanf("%d", &n); } while(n <= 0);
    do {
        printf("\n1. Check whether all digits are even\n");
        printf("2. Check whether all digits are prime\n3. Exit\nEnter choice: ");
        scanf("%d", &choice);
        switch(choice) {
            case 1:
                temp = n; allEven = 1;
                while(temp > 0) { d = temp % 10; if(d % 2 != 0) allEven = 0; temp /= 10; }
                if(allEven) printf("All digits are even\n");
                else printf("All digits not even\n"); break;
            case 2:
                temp = n; allPrime = 1;
                while(temp > 0) {
                    d = temp % 10;
                    if(!(d == 2 || d == 3 || d == 5 || d == 7)) allPrime = 0;
                    temp /= 10;
                }
                if(allPrime) printf("All digits are prime\n");
                else printf("All digits are not prime\n"); break;
            case 3: break;
            default: printf("Invalid choice.\n");
        }
    } while(choice != 3);
    return 0;
}
```

Compilation, Execution & Testing

```
puppie@Anvikas-MacBook-Air Desktop % clang day15.c -o day15
puppie@Anvikas-MacBook-Air Desktop % ./day15
Enter a positive integer: 12345
Enter choice: 1
All digits not even
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day15
Enter a positive integer: 24680
Enter choice: 1
All digits are even
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day15
Enter a positive integer: 2357
Enter choice: 2
All digits are prime
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day15
Enter a positive integer: 2379
Enter choice: 2
All digits are not prime
```

Result: The menu-driven C program was compiled, executed and tested successfully against the prepared test cases.